



CPU & GPU

The heart and soul of your PC

CPU

AVX Instructions

AVX (Advanced Vector Extensions) is a set of instructions that improve CPU performance in tasks like digital signal processing, 3D modeling, and video encoding. AVX and its successors (AVX2, AVX-512) allow CPUs to handle more data in parallel, speeding up specific workloads. Enthusiasts using applications that benefit from AVX ins

Base Clock (BCLK)

The base clock (BCLK) is the foundational clock speed of the CPU, from which other frequencies (like core clock and memory clock) are derived. Overclockers sometimes adjust the BCLK in addition to the multiplier to fine-tune performance. However, tweaking the BCLK can also affect other system components, so it's tricky.

Boost Clock

Boost clock refers to the maximum clock speed the CPU can reach under load, thanks to dynamic overclocking technologies like Intel Turbo Boost or AMD Precision Boost. The CPU automatically increases its frequency when thermal and power conditions allow, providing extra performance in demanding tasks.

Branch Prediction

Branch prediction is a technique used by modern CPUs to improve instruction execution efficiency by guessing the path of a conditional operation before it's fully processed. This reduces delays and improves throughput. Enthusiasts interested in CPU architecture study the sophistication of branch prediction algorithms, as better prediction reduces pipeline stalls and increases overall CPU performance in real-time computing.